Keras Traffic Prediction

with ComNetsEmu

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INTRODUCTION

Mininet:

- Network topology
- Traffic generation

Ryu controller:

- Switches
- Logging
- CSV

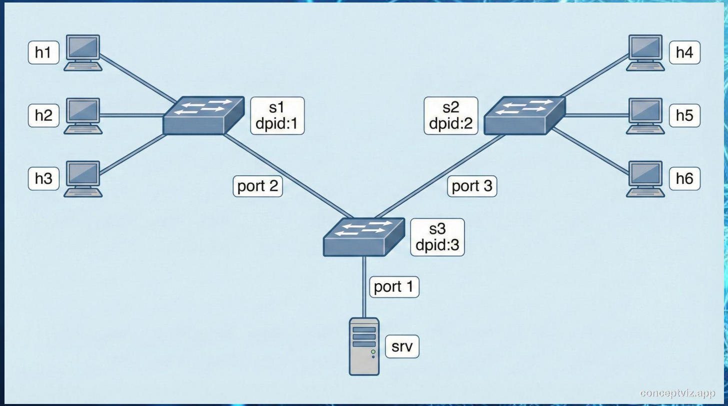
Keras:

- Model
- LSTM
- Output

Conclusions:

- Improvements
- Ideas

NETWORK - TRAFFIC



Network topology:

- 3 switch topology with central server: single observation point on switch 3.
- Hosts distributed across edge switches: realistic multi-client simulation.
- Parameterized links (bw/delay/loss): control over network conditions.

Network Traffic:

- UDP + HTTP traffic to combine continuous flows and application bursts.
- Random scenario: background noise, random bursts and random HTTP requests.
- Cyclical scenario: repeating phases of low/medium/high traffic followed by cooldown.

RYU CONTROLLER

Forwarding:

- L2 learning switch: learns MAC/port mappings and reduces PacketIn events over time.
- Table-miss: unmatched packets sent to the controller.

Logging:

- Periodic PortStats polling: sampling at fixed intervals.
- Per-port delta saved to CSV: time series ready for prediction.

MODEL

Input:

- Input layer is a (batch, x, 12) tensor where x is the sequence length and 12 is the ports data with format [p1rx_bytes, p1tx_bytes, p1rx_packets, p1tx_packets … (other ports)].

LSTM:

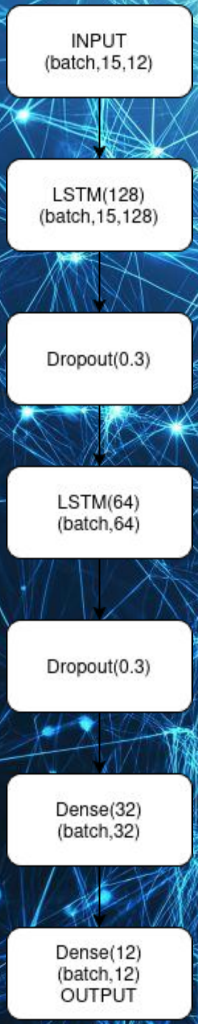
- Long-Short-Term-Memory layer which elaborates sequences of data and keeps a memory state of past data.

Dropout:

- Randomly zeroes values for better adaptability (training only).

Dense:

- Map features to the desired output size.



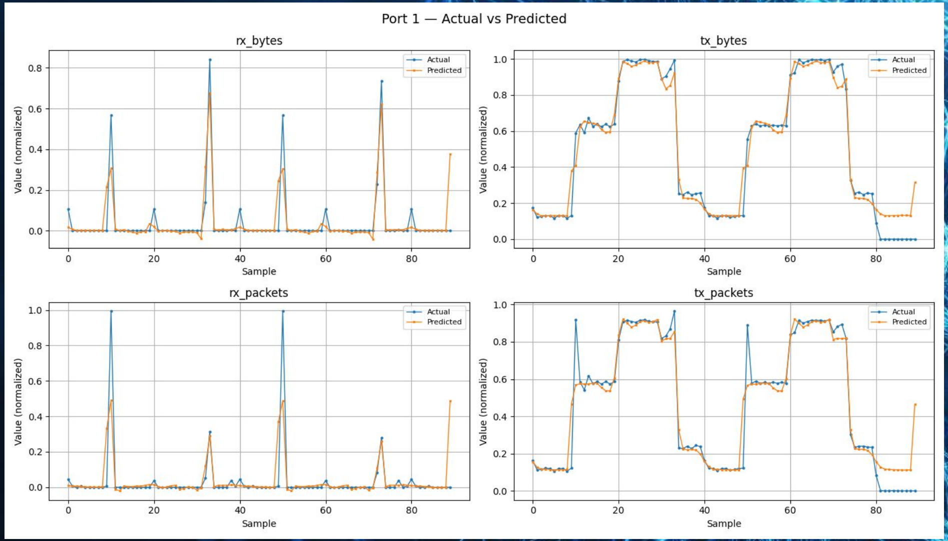
OUTPUT - PLOTS

Output:

- Outputs of the model are a single tensor with shape (12) predicting the next interval data on the switch.

Plots:

- This plot on the sides demonstrates a possible model inference on a test set for a cyclic, recurrent traffic pattern.



Problems:

- Obviously hard to predict randomized or “spiky” traffic patterns and also easily overfits the dataset.

CONCLUSIONS

Improvements:

- More data, bigger sequences for performance.
- Model structure, loss functions and data handling.

Possible tweaks:

- Implement continuous training parallel to the network.
- Convert model results into a congestion control predictor.

Real use cases:

- Switch performance analysis.
- Resources administration.